EAS 550 – Infrastructure Provisioning, Relational Modeling & Data Ingestion

Perishable Inventory Intelligence Platform

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I. Dataset Overview

This project is using the Managing Perishable Inventory Data dataset, which includes transaction records of perishble goods across various stores, suppliers and regions. It contain details such as product information, supplier performance, pricing, demand and also data related to expiration.

The main objective of this project is to convert the raw dataset into a structured relational database, which reduces redundancy and help in efficient querying for future analytics and decision making process.

II. Entity and Relationship Design

After analyzing the dataset, we identified the following core entities:

- **Regions**: Geographical groupings of stores.
- **Stores**: Retail locations where products are sold.
- **Categories**: Product classifications or types.
- **Products**: Individual inventory items.
- **Suppliers**: Entities that provide products.
- **Promotions**: Discounts or offers applied to products.
- **Inventory Transactions**: The central fact table representing daily inventory activity.

The relationships between these entities are as follows:

- One **Region** has many **Stores**.
- One **Category** has many **Products**.
- One **Store**, **Product**, and **Supplier** are linked through **Inventory Transactions**.
- **Products** and **Promotions** have a many-to-many relationship.

This structure help to improve clarity and scalability by separating the descriptive entities from the transactional data. Figure 1 show the Entity Relationship Diagram (ERD) for the proposed database schema.

III. Normalization (1NF to 3NF)

The schema was created to follow Third Normal Form (3NF) so that redundancy is reduced and data integrity is maintained.

A. First Normal Form (1NF)

All the attributes have atomic and single-valued data. In preprocessing step, missing values was handled, wrong data types were corrected, and duplicate records were removed. Because of this, the dataset become suitable for storing in relational format.

B. Second Normal Form (2NF)

The original dataset had a composite transactional structure which includes product, store, supplier and date. The attributes which were only depending on part of this structure was separated into different tables:

- Product-specific attributes were moved to the **Products** table.
- Supplier-specific attributes were moved to the **Suppliers** table.
- Store and region-related attributes were moved to the **Stores** and **Regions** tables.

This separation make sure that non-key attributes are fully dependent on the complete key in their respective tables.

C. Third Normal Form (3NF)

Transitive dependencies were handled by making sure that all non-key attributes depend only on the primary key in their respective tables.

- Region names are stored only in the **Regions** table.
- Category names are stored only in the **Categories** table.
- Supplier scores are stored only in the **Suppliers** table.

This helps each table to represent a single idea and also reduces redundant data storage.

IV. Many-to-Many Relationship Resolution

There is a many-to-many relationship between Products and Promotions, since one product can belong to multiple promotions at different times, and one promotion can be linked to many products.

This relationship was resolved using a bridge table named **product_promotions**, which contains:

- **product_id**
- **promotion_id**
- **valid_from**

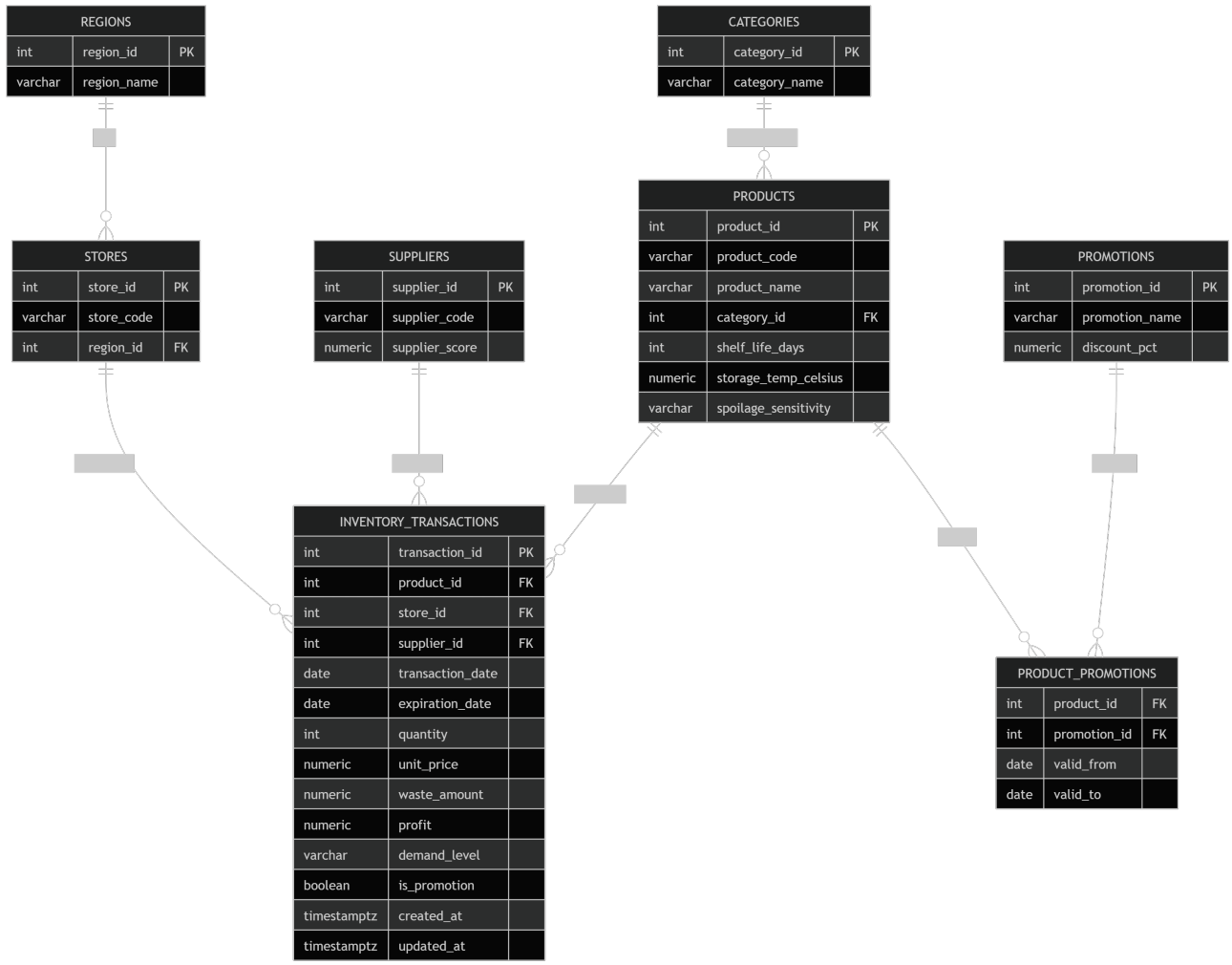

Fig. 1. Entity Relationship Diagram for the Perishable Inventory Intelligence Platform.

- **valid_to**

This design preserves relationship history while maintaining normalization.

V. Anomaly Prevention

The schema design helps prevent common database anomalies:

- **Insertion Anomaly**: New products, suppliers, or stores can be added without requiring a transaction record.
- **Update Anomaly**: Changes such as updating a supplier score or region name are made in only one place, preventing inconsistencies.
- **Deletion Anomaly**: Deleting a transaction record does not remove important reference data such as product, supplier, or store details.

VI. Summary

The final schema is normalized up to 3NF, where many-to-many relationships are solved using bridge tables, and data integrity is maintained by properly separating entities. This structure gives a good base for scalable data ingestion, storage, and future analytical tasks in later phases of the project.